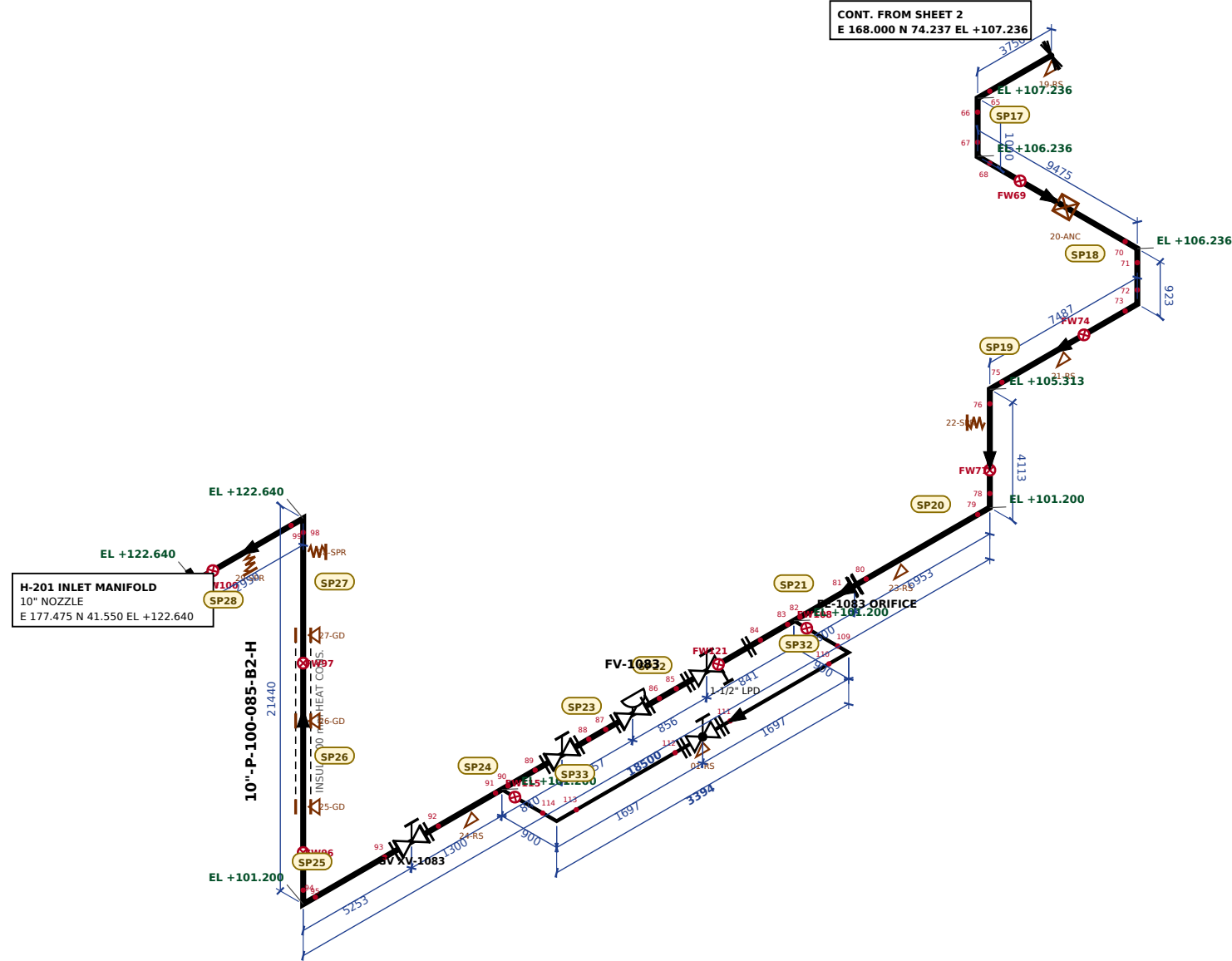
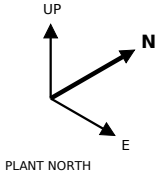


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 10"-P-100-085-B2-H	FROM P-112A/B	TO H-201 inlet manifold	PIPING CLASS B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T 24.0 barg / 380.0 C	OPER. P / T 18.99 barg / 348.7 C	TEST P (HYDRO) 44.7 barg (1.5xPxST/5=1.24)	PIPE WALL SCH 40 (9.27 mm)
INSULATION H 100 mm	PWHT YES - all thk (P-No.5B, B31.3 Tbl	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; rotating equipment (
P&ID CFU-100-PR-PID-006, CFU-200-PR-PID-012	FLUID / PHASE P / L Atmospheric residue to H	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 33 spools; 89 shop / 32 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE SCH 40, BE, ASTM A335 P5 SMLS	10"	60.4 m
2	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	10"	8
3	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5 (orifice)	10"	2
4	TEE RED., BW, SCH 40, ASTM A234 WP5	10" x 6"	2
5	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	10"	9
6	PIPE SCH 40, BE, ASTM A335 P5 SMLS	6"	3.2 m
7	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	6"	2
8	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	6"	2
9	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	1
10	ERECTION: ORIFICE PLATE + ORIFICE FLG PAIR FE-1083 (by I&C)	10"	1
11	GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	10"	3
12	CONTROL VALVE FV-1083 CL300 RF (by I&C)	10"	1
13	GLOBE VALVE CL300 RF, A217 C5, trim 5	6"	1
14	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	10"	11
15	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	10"	11
16	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	6"	2
17	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	6"	2
18	PIPE SUPPORTS (std. dwgs): anchor x1, guide x3, rest x5, spring x3	-	12

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP17-1	10"	4488	SP17-2	10"	238	SP17-3	10"	2250
SP18-1	10"	6463	SP18-2	10"	161	SP18-3	10"	2250
SP19-1	10"	4475	SP19-2	10"	2250	SP20-1	10"	1102
SP20-2	10"	6456	SP21-1	10"	1268	SP21-2	10"	280
SP22-1	10"	20	SP23-1	10"	20	SP24-1	10"	280
SP24-2	10"	739	SP25-1	10"	4527	SP25-2	10"	2250
SP26-1	10"	11700	SP27-1	10"	6728	SP27-2	10"	2250
SP28-1	10"	203	SP32-1	6"	455	SP32-2	6"	1149
SP33-1	6"	1149	SP33-2	6"	455			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP17=8.5m SP18=10.4m SP19=7.5m SP20=8.4m SP21=2.2m SP22=0.3m SP23=0.3m SP24=1.7m SP25=7.7m SP26=11.7m
SP27=9.7m SP28=0.3m SP32=2.2m SP33=2.2m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 44.7 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Discharge check + block valves in riser above each pump (accessible from grade).
- Rack run on tier 1 (hot B2) at rack south edge; expansion loops raised +1.0 m.
- FV-1083 / FE / XV station at grade (EL 101.2) north of H-201 for operator access.
- Riser at H-201 supported from heater structure with variable spring hangers.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 10"-P-100-085-B2-H P-112A/B DISCHARGE TO H-201 INLET		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DRAWING NUMBER CFU-100-PI-ISO-004	DATE 2026-10-02	REV A